

Azure Virtual Networking: Theory & Implementation Guide

Table of Contents

- 1. [Introduction to Azure Virtual Networking](#)
 - 2. [Core Concepts & Theory](#)
 - 3. [Network Architecture Design](#)
 - 4. [Azure Bastion Host](#)
 - 5. [Implementation Steps](#)
 - 6. [Architecture Diagram](#)
 - 7. [Best Practices](#)
 - 8. [Summary](#)
-

1. Introduction to Azure Virtual Networking

Azure Virtual Networking is the **fundamental building block** for private networks in Microsoft Azure. It enables Azure resources like Virtual Machines (VMs), databases, and web apps to securely communicate with each other, the internet, and on-premises networks.

Why Virtual Networking?

- Isolate and protect cloud resources
 - Control inbound and outbound traffic
 - Enable hybrid cloud connectivity
 - Implement zero-trust security model
-

2. Core Concepts & Theory

2.1 Virtual Network (VNet)

A **VNet** is a logically isolated network in Azure that closely resembles a traditional on-premises network.

Property	Description
Address Space	CIDR block assigned to the VNet (e.g., 10.0.0.0/16)
Region	VNet is bound to a specific Azure region
Subscription	VNet belongs to a single Azure subscription
Isolation	Complete isolation from other VNets by default

2.2 Subnets

Subnets **divide a VNet** into smaller, manageable segments.

```
VNet: 10.0.0.0/16
├─ Subnet-Web:      10.0.1.0/24   (Public-facing)
├─ Subnet-App:      10.0.2.0/24   (Application layer)
├─ Subnet-DB:       10.0.3.0/24   (Database layer)
└─ AzureBastionSubnet: 10.0.4.0/27 (Bastion - Reserved Name)
```

Key Subnet Facts:

- Azure reserves **5 IP addresses** per subnet (first 4 + last 1)
- Subnets in same VNet can communicate by default
- Each subnet can have its own Network Security Group (NSG)

2.3 Network Security Groups (NSG)

NSGs act as a **virtual firewall** controlling traffic to/from Azure resources.

NSG Rule Components:

Component	Description	Example
Priority	Lower number = higher priority	100, 200, 300
Source	Origin of traffic	IP, CIDR, Service Tag
Destination	Target of traffic	IP, CIDR, Any
Protocol	Traffic protocol	TCP, UDP, ICMP, Any
Port Range	Port or range	80, 443, 3389, 22
Action	Allow or Deny	Allow / Deny

Default NSG Rules (Inbound):

Priority	Name	Action
65000	AllowVNetInBound	Allow
65001	AllowAzureLoadBalancerInBound	Allow
65500	DenyAllInBound	Deny

2.4 Public vs Private IP Addresses

Type	Scope	Use Case	Persistence
Public IP (Dynamic)	Internet-facing	Dev/Test VMs	Changes on stop/start
Public IP (Static)	Internet-facing	Production, DNS	Never changes
Private IP (Dynamic)	Internal VNet	Backend servers	May change
Private IP (Static)	Internal VNet	Domain controllers, DBs	Fixed

2.5 Azure DNS

- Each VNet gets an **Azure-provided DNS** automatically
- Default DNS: **168.63.129.16**
- Custom DNS servers can be configured
- VM hostnames resolvable within VNet

2.6 VNet Peering

Connects **two VNets** allowing resources to communicate as if in same network.

```
VNet-A (10.0.0.0/16) <----Peering----> VNet-B (10.1.0.0/16)
```

Types:

- **Regional Peering** — Same Azure region
- **Global Peering** — Different Azure regions

2.7 Network Routing

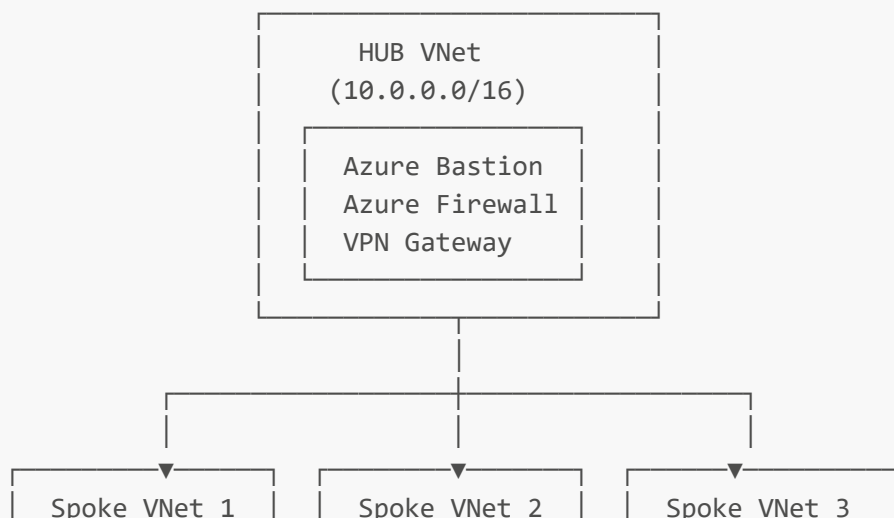
Azure automatically creates **system routes** for traffic flow.

Route Table Example:

Destination	Next Hop Type
0.0.0.0/0	Internet
10.0.0.0/16	VirtualNetwork
10.0.1.0/24	VirtualNetwork

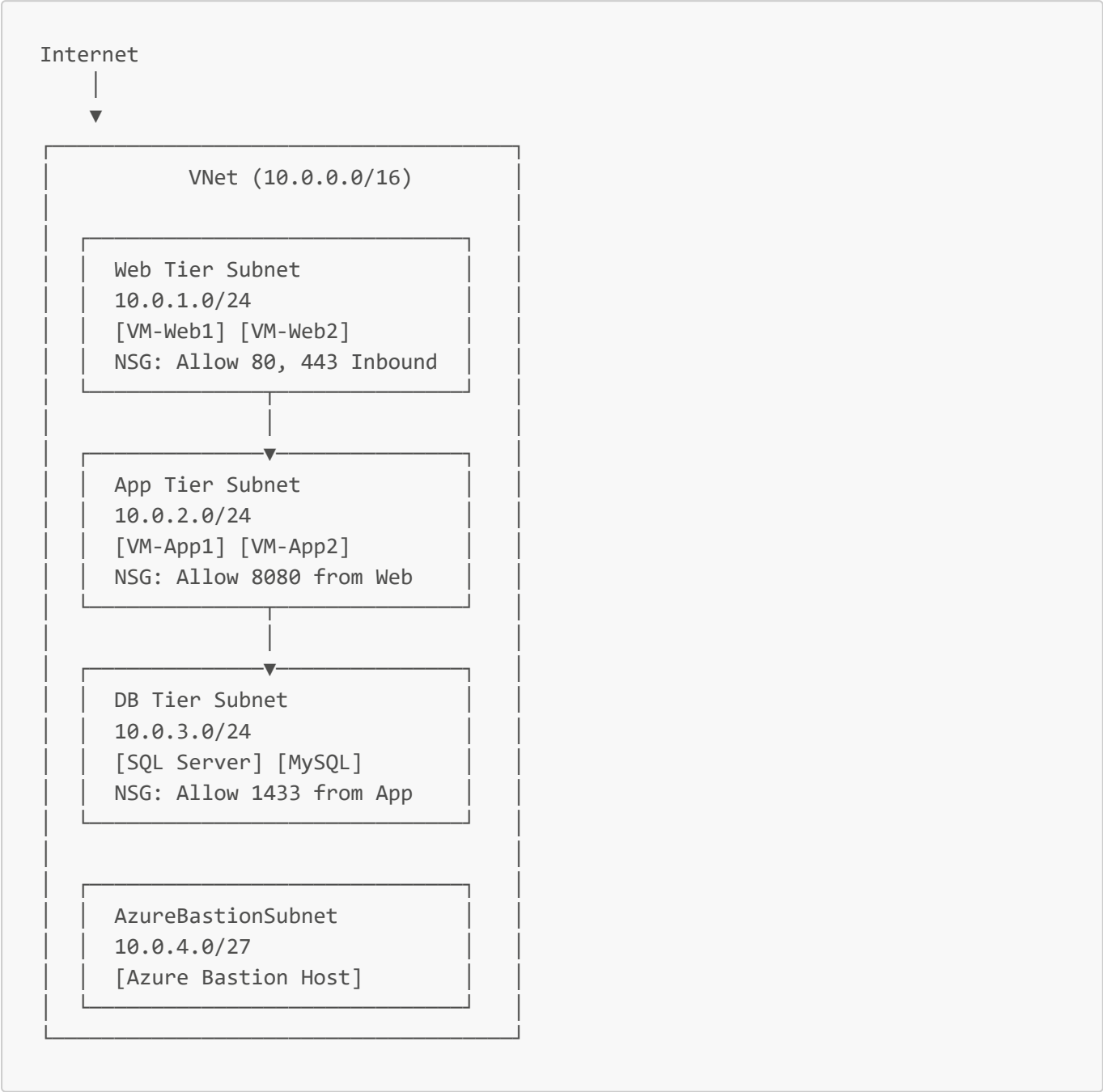
3. Network Architecture Design

3.1 Hub-Spoke Architecture (Recommended)



(Production) 10.1.0.0/16	(Development) 10.2.0.0/16	(Testing) 10.3.0.0/16
-----------------------------	------------------------------	--------------------------

3.2 Three-Tier Architecture (Common Pattern)



3.3 IP Address Planning

Organization VNet Plan:			
└─ VNet Address Space: 10.0.0.0/8 (Large Enterprise)			
└─ Production: 10.1.0.0/16			
└─	Web:	10.1.1.0/24	→ 251 usable hosts
└─	App:	10.1.2.0/24	→ 251 usable hosts
└─	DB:	10.1.3.0/24	→ 251 usable hosts

```
├─ Development: 10.2.0.0/16
│   └─ Web:      10.2.1.0/24
│     └─ App:     10.2.2.0/24
└─ Bastion:      10.1.4.0/27 → 27 usable hosts (min /27 for Bastion)
```

4. Azure Bastion Host

4.1 What is Azure Bastion?

Azure Bastion is a **fully managed PaaS service** that provides secure and seamless RDP/SSH connectivity to VMs **directly from the Azure portal** over TLS (port 443) — without exposing VMs to the public internet.

4.2 Traditional vs Bastion Access

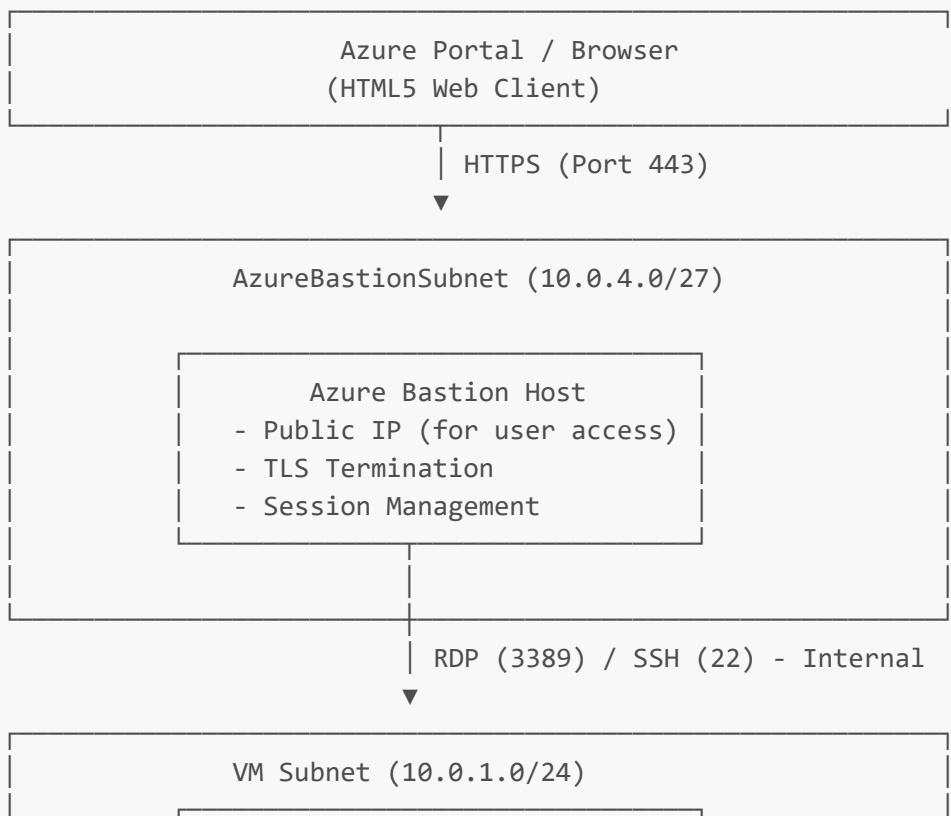
✗ TRADITIONAL (Insecure):

User —(Public Internet)→ VM Public IP → RDP/SSH Port Open
(Attack Surface!)

✓ WITH BASTION (Secure):

User —(HTTPS/443)→ Azure Bastion → VM Private IP → RDP/SSH
(Managed PaaS) (No Public IP on VM!)

4.3 How Azure Bastion Works



```
Virtual Machine
Private IP: 10.0.1.4
No Public IP Required! ✓
```

4.4 Azure Bastion SKUs

Feature	Basic SKU	Standard SKU
RDP/SSH via Portal	✓	✓
Custom Ports	✗	✓
Native Client Support	✗	✓
IP-based Connection	✗	✓
Shareable Links	✗	✓
File Transfer	✗	✓
Scale Units	2	2–50
Price	Lower	Higher

4.5 Bastion Security Benefits

- ✓ No RDP/SSH ports exposed to internet
- ✓ No need for Public IPs on VMs
- ✓ Protection against port scanning
- ✓ Zero-day RDP/SSH vulnerability protection
- ✓ Session recording & audit logs (Standard)
- ✓ MFA support through Azure AD
- ✓ SSL/TLS encrypted sessions

4.6 Required NSG Rules for Bastion

Bastion Subnet NSG — Inbound Rules:

Priority	Source	Port	Destination	Action
100	Internet	443	Bastion	Allow
110	GatewayManager	443	Bastion	Allow ← Required!
120	AzureLoadBalancer	443	Bastion	Allow ← Required!

Bastion Subnet NSG — Outbound Rules:

Priority	Source	Port	Destination	Action
100	Bastion	3389, 22	VirtualNetwork	Allow ← RDP/SSH to VMs

110	Bastion	443	AzureCloud	Allow	← Azure services
-----	---------	-----	------------	-------	------------------

5. Implementation Steps

Prerequisites

- Active Azure Subscription
 - Azure Portal access or Azure CLI installed
 - Contributor or Owner role on the subscription
-

Step 1: Create a Resource Group

Via Azure Portal:

1. Login → portal.azure.com
2. Search → "Resource Groups"
3. Click → "+ Create"
4. Fill Details:
 - Subscription: Your Subscription
 - Resource Group Name: RG-NetworkDemo
 - Region: East US (or your preferred region)
5. Click → "Review + Create" → "Create"

Via Azure CLI:

```
# Login to Azure
az login

# Create Resource Group
az group create \
  --name RG-NetworkDemo \
  --location eastus

# Verify creation
az group show --name RG-NetworkDemo
```

Step 2: Create Virtual Network with Subnets

Via Azure Portal:

1. Search → "Virtual Networks"
2. Click → "+ Create"

3. Basics Tab:
 - Resource Group: RG-NetworkDemo
 - Name: VNet-Main
 - Region: East US
4. IP Addresses Tab:
 - IPv4 Address Space: 10.0.0.0/16
 - Delete default subnet
 - Add Subnet → Web-Subnet
 - └─ Address Range: 10.0.1.0/24
 - Add Subnet → App-Subnet
 - └─ Address Range: 10.0.2.0/24
 - Add Subnet → DB-Subnet
 - └─ Address Range: 10.0.3.0/24
 - Add Subnet → AzureBastionSubnet ← Exact name required!
 - └─ Address Range: 10.0.4.0/27
5. Click → "Review + Create" → "Create"

Via Azure CLI:

```
# Create Virtual Network
az network vnet create \
  --resource-group RG-NetworkDemo \
  --name VNet-Main \
  --address-prefix 10.0.0.0/16 \
  --location eastus

# Create Web Subnet
az network vnet subnet create \
  --resource-group RG-NetworkDemo \
  --vnet-name VNet-Main \
  --name Web-Subnet \
  --address-prefix 10.0.1.0/24

# Create App Subnet
az network vnet subnet create \
  --resource-group RG-NetworkDemo \
  --vnet-name VNet-Main \
  --name App-Subnet \
  --address-prefix 10.0.2.0/24

# Create DB Subnet
az network vnet subnet create \
  --resource-group RG-NetworkDemo \
  --vnet-name VNet-Main \
  --name DB-Subnet \
  --address-prefix 10.0.3.0/24

# Create Bastion Subnet (Name must be exactly AzureBastionSubnet)
az network vnet subnet create \
  --resource-group RG-NetworkDemo \
  --vnet-name VNet-Main \
```

```
--name AzureBastionSubnet \  
--address-prefix 10.0.4.0/27  
  
# Verify VNet and Subnets  
az network vnet show \  
  --resource-group RG-NetworkDemo \  
  --name VNet-Main \  
  --output table
```

Step 3: Create Network Security Groups (NSG)

Create NSG for Web Tier:

```
# Create Web NSG  
az network nsg create \  
  --resource-group RG-NetworkDemo \  
  --name NSG-Web \  
  --location eastus  
  
# Allow HTTP (Port 80)  
az network nsg rule create \  
  --resource-group RG-NetworkDemo \  
  --nsg-name NSG-Web \  
  --name Allow-HTTP \  
  --priority 100 \  
  --protocol Tcp \  
  --destination-port-range 80 \  
  --access Allow \  
  --direction Inbound  
  
# Allow HTTPS (Port 443)  
az network nsg rule create \  
  --resource-group RG-NetworkDemo \  
  --nsg-name NSG-Web \  
  --name Allow-HTTPS \  
  --priority 110 \  
  --protocol Tcp \  
  --destination-port-range 443 \  
  --access Allow \  
  --direction Inbound  
  
# Deny all other Inbound  
az network nsg rule create \  
  --resource-group RG-NetworkDemo \  
  --nsg-name NSG-Web \  
  --name Deny-All-Inbound \  
  --priority 4096 \  
  --protocol '*' \  
  --destination-port-range '*' \
```

```
--access Deny \  
--direction Inbound
```

Create NSG for DB Tier:

```
# Create DB NSG  
az network nsg create \  
  --resource-group RG-NetworkDemo \  
  --name NSG-DB \  
  --location eastus  
  
# Allow SQL from App Subnet only  
az network nsg rule create \  
  --resource-group RG-NetworkDemo \  
  --nsg-name NSG-DB \  
  --name Allow-SQL-From-App \  
  --priority 100 \  
  --protocol Tcp \  
  --source-address-prefix 10.0.2.0/24 \  
  --destination-port-range 1433 \  
  --access Allow \  
  --direction Inbound  
  
# Deny all other Inbound  
az network nsg rule create \  
  --resource-group RG-NetworkDemo \  
  --nsg-name NSG-DB \  
  --name Deny-All \  
  --priority 4096 \  
  --protocol '*' \  
  --destination-port-range '*' \  
  --access Deny \  
  --direction Inbound
```

Associate NSGs to Subnets:

```
# Associate Web NSG to Web Subnet  
az network vnet subnet update \  
  --resource-group RG-NetworkDemo \  
  --vnet-name VNet-Main \  
  --name Web-Subnet \  
  --network-security-group NSG-Web  
  
# Associate DB NSG to DB Subnet  
az network vnet subnet update \  
  --resource-group RG-NetworkDemo \  
  --vnet-name VNet-Main \  
  --name DB-Subnet
```

```
--name DB-Subnet \  
--network-security-group NSG-DB
```

Step 4: Create Virtual Machine (Without Public IP)

Via Azure Portal:

1. Search → "Virtual Machines"
2. Click → "+ Create" → "Azure Virtual Machine"
3. Basics Tab:
 - Resource Group: RG-NetworkDemo
 - VM Name: VM-Web01
 - Region: East US
 - Image: Windows Server 2022 / Ubuntu 22.04
 - Size: Standard_B2s
 - Username: azureadmin
 - Password: [Strong Password]
4. Networking Tab:
 - Virtual Network: VNet-Main
 - Subnet: Web-Subnet (10.0.1.0/24)
 - Public IP: NONE ← Important for Bastion!
 - NIC NSG: None (we created custom NSG)
5. Click → "Review + Create" → "Create"

Via Azure CLI:

```
# Create VM without Public IP  
az vm create \  
  --resource-group RG-NetworkDemo \  
  --name VM-Web01 \  
  --image Ubuntu2204 \  
  --vnet-name VNet-Main \  
  --subnet Web-Subnet \  
  --public-ip-address "" \  
  --admin-username azureadmin \  
  --admin-password "SecurePassword123!" \  
  --size Standard_B2s \  
  --location eastus  
  
# Verify VM - should show no public IP  
az vm show \  
  --resource-group RG-NetworkDemo \  
  --name VM-Web01 \  
  --show-details \  
  --
```

```
--query "{Name:name, PrivateIP:privateIps, PublicIP:publicIps}" \  
--output table
```

Step 5: Deploy Azure Bastion Host

Via Azure Portal:

1. Search → "Bastions"
2. Click → "+ Create"
3. Basics Tab:
 - Resource Group: RG-NetworkDemo
 - Name: Bastion-Main
 - Region: East US (Same as VNet!)
 - Tier: Basic (or Standard)
 - Virtual Network: VNet-Main
 - Subnet: AzureBastionSubnet (auto-detected)
 - Public IP Address: Create New
 - └─ Name: Bastion-PublicIP
4. Click → "Review + Create" → "Create"
⚠ Deployment takes 5-10 minutes

Via Azure CLI:

```
# Create Public IP for Bastion  
az network public-ip create \  
  --resource-group RG-NetworkDemo \  
  --name Bastion-PublicIP \  
  --sku Standard \  
  --location eastus  
  
# Create Azure Bastion  
az network bastion create \  
  --resource-group RG-NetworkDemo \  
  --name Bastion-Main \  
  --public-ip-address Bastion-PublicIP \  
  --vnet-name VNet-Main \  
  --location eastus \  
  --sku Basic  
  
# Check Bastion deployment status  
az network bastion show \  
  --resource-group RG-NetworkDemo \  
  --name Bastion-Main \  
  --query "provisioningState"
```

Step 6: Connect to VM via Azure Bastion

Via Azure Portal:

1. Navigate to → Virtual Machines → VM-Web01
2. Click → "Connect" in the top menu
3. Select → "Bastion" tab
4. Click → "Use Bastion"
5. Enter Credentials:
 - Username: azureadmin
 - Password: [Your Password]
6. Click → "Connect"
 - ✅ Browser opens RDP/SSH session securely!

Via Azure CLI (Standard SKU only):

```
# SSH via Bastion using native client
az network bastion ssh \
  --name Bastion-Main \
  --resource-group RG-NetworkDemo \
  --target-resource-id $(az vm show -g RG-NetworkDemo -n VM-Web01 --query id -o tsv) \
  --auth-type password \
  --username azureadmin

# RDP via Bastion (Windows VM)
az network bastion rdp \
  --name Bastion-Main \
  --resource-group RG-NetworkDemo \
  --target-resource-id $(az vm show -g RG-NetworkDemo -n VM-Web01 --query id -o tsv)
```

Step 7: Verify Network Configuration

```
# List all resources in Resource Group
az resource list \
  --resource-group RG-NetworkDemo \
  --output table

# Check VNet details
az network vnet show \
  --resource-group RG-NetworkDemo \
  --name VNet-Main

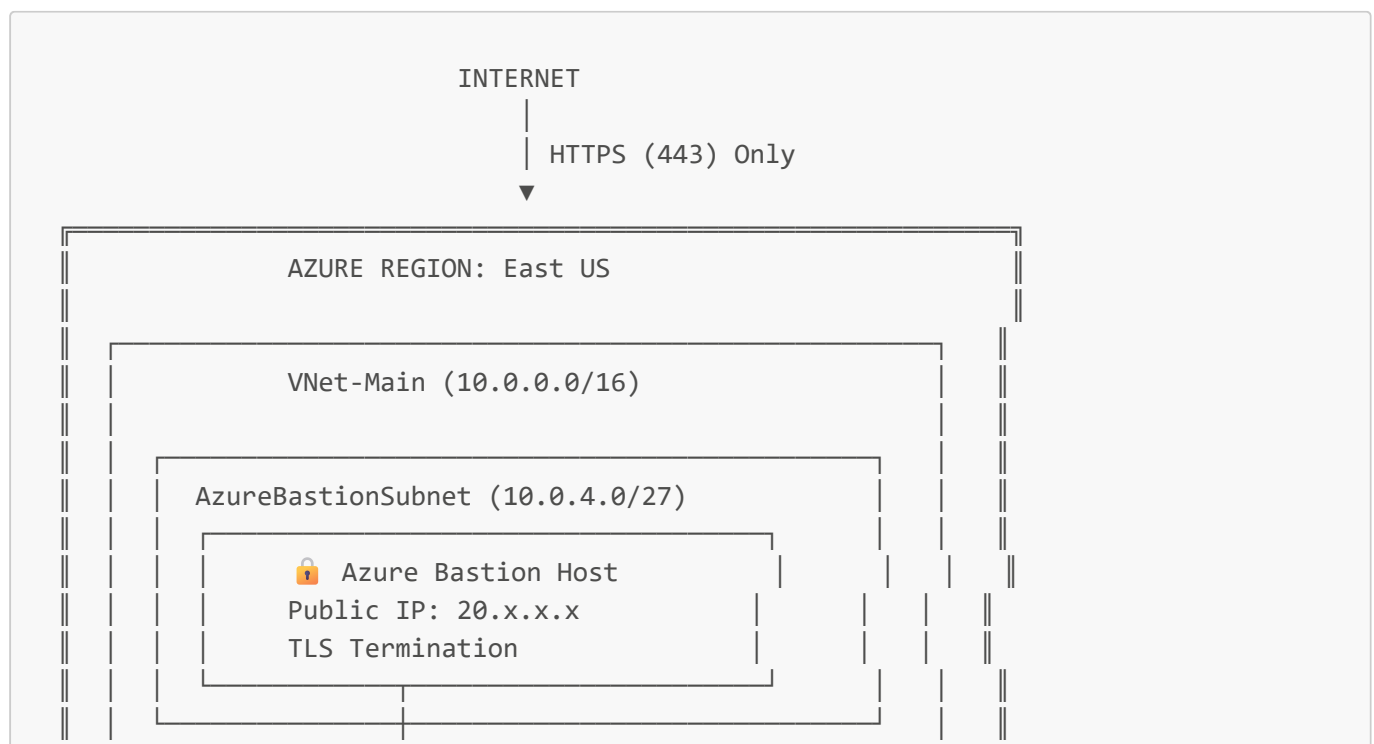
# List all subnets
az network vnet subnet list \
```

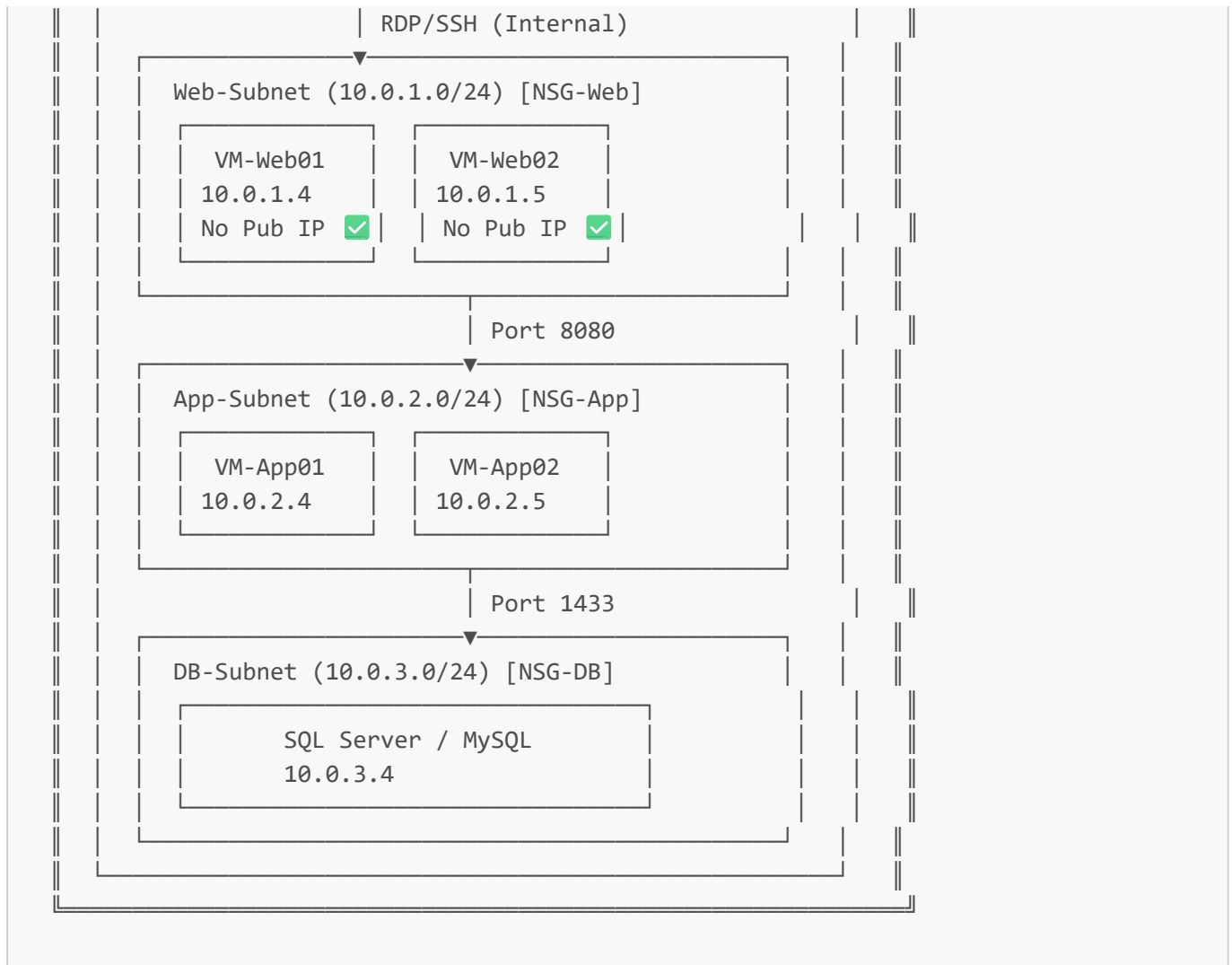
```
--resource-group RG-NetworkDemo \  
--vnet-name VNet-Main \  
--output table  
  
# Check NSG rules  
az network nsg rule list \  
  --resource-group RG-NetworkDemo \  
  --nsg-name NSG-Web \  
  --output table  
  
# Check effective security rules on VM NIC  
az network nic show-effective-nsg \  
  --resource-group RG-NetworkDemo \  
  --name VM-Web01VMNic \  
  --output table
```

Step 8: Cleanup Resources (Optional)

```
# Delete entire Resource Group (removes all resources)  
az group delete \  
  --name RG-NetworkDemo \  
  --yes \  
  --no-wait  
  
# Verify deletion  
az group exists --name RG-NetworkDemo
```

6. Architecture Diagram





7. Best Practices

7.1 VNet Design

- ✅ DO:
 - Plan IP address space before deployment
 - Use non-overlapping CIDR blocks
 - Leave room for future growth
 - Use separate subnets for each tier
 - Enable DDoS Protection for production
- ❌ DON'T:
 - Use public IP ranges (1.0.0.0/8) for VNets
 - Create VNets too small (/29 or smaller)
 - Mix workloads in same subnet without NSG
 - Expose RDP/SSH ports to internet

7.2 NSG Best Practices

- ✅ DO:
 - Apply NSGs at subnet level (not just NIC)
 - Use Service Tags instead of IP ranges
 - Document all NSG rules
 - Use Deny-All as last rule
 - Regular audit of NSG rules
- ✅ Service Tags to Use:
 - Internet → All internet traffic
 - VirtualNetwork → All VNet traffic
 - AzureCloud → Azure datacenter IPs
 - GatewayManager → Azure Gateway manager

7.3 Azure Bastion Best Practices

- ✅ DO:
 - Use Standard SKU for production
 - Enable diagnostic logs
 - Use /27 or larger for BastionSubnet
 - Enable MFA for Azure portal access
 - Regularly review Bastion access logs
- ❌ DON'T:
 - Deploy other resources in AzureBastionSubnet
 - Remove required NSG rules for Bastion
 - Use Basic SKU for enterprise deployments

7.4 Security Checklist

Item	Status
VMs have no public IP	✅
NSGs applied to all subnets	✅
RDP/SSH not open to internet	✅
Azure Bastion deployed	✅
DDoS Protection enabled	✅
Network Watcher enabled	✅
Diagnostic logs configured	✅
VNet Flow Logs enabled	✅

8. Summary

What We Implemented:

COMPONENT	RESOURCE NAME	VALUE
Resource Group	→ RG-NetworkDemo	
Virtual Network	→ VNet-Main	10.0.0.0/16
Web Subnet	→ Web-Subnet	10.0.1.0/24
App Subnet	→ App-Subnet	10.0.2.0/24
DB Subnet	→ DB-Subnet	10.0.3.0/24
Bastion Subnet	→ AzureBastionSubnet	10.0.4.0/27
Web NSG	→ NSG-Web	Allow 80, 443
DB NSG	→ NSG-DB	Allow 1433 from App
Virtual Machine	→ VM-Web01	No Public IP
Bastion Host	→ Bastion-Main	Secure RDP/SSH

Key Takeaways:

- 1. **VNet** provides network isolation in Azure
- 2. **Subnets** segment the network for security tiers
- 3. **NSGs** act as virtual firewalls with priority-based rules
- 4. **Azure Bastion** eliminates need for public IPs on VMs
- 5. **Zero Public Exposure** — VMs accessible only via Bastion
- 6. **Defense in Depth** — Multiple security layers implemented

References

Resource	Link
Azure VNet Documentation	docs.microsoft.com/azure/virtual-network
Azure Bastion Documentation	docs.microsoft.com/azure/bastion
NSG Documentation	docs.microsoft.com/azure/network-security-groups
Azure CLI Reference	docs.microsoft.com/cli/azure
Azure Pricing Calculator	azure.microsoft.com/pricing/calculator

Documentation prepared for Azure Virtual Networking Lab
Last Updated: 2024
Author: Azure Network Implementation Guide

